



A systematic study of some promising electrolyte additives in $\text{Li}[\text{Ni}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}]\text{O}_2/\text{graphite}$, $\text{Li}[\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}]/\text{graphite}$ and $\text{Li}[\text{Ni}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}]/\text{graphite}$ pouch cells

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HIGHLIGHTS

- Advanced additives were compared to vinylene carbonate in NMC111, NMC532 and NMC622 cells.
- At 4.2 V, all advanced additives performed well with all positive electrode materials.
- At 4.4 V, NMC622 shows more gas production than the other materials with all additives.

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ABSTRACT

$\text{Li}[\text{Ni}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}]\text{O}_2/\text{graphite}$, $\text{Li}[\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}]\text{O}_2/\text{graphite}$ and $\text{Li}[\text{Ni}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}]/\text{graphite}$ pouch cells were examined with and without electrolyte additives using the ultra high precision charger at Dalhousie University, electrochemical impedance spectroscopy, gas evolution measurements and “cycle-store” tests. The electrolyte additives tested were vinylene carbonate (VC), prop-1-ene-1,3-sultone (PES), pyridine-boron trifluoride (PBF), 2% PES + 1% methylene methanedisulfonate (MMDS) + 1% tris(trimethylsilyl) phosphite (TTSPi) and 0.5% pyrazine di-boron trifluoride (PRZ) + 1% MMDS. The charge end-point capacity slippage, capacity fade, coulombic efficiency, impedance change during cycling, gas evolution and voltage drop during “cycle-store” testing were compared to gain an understanding of the effects of these promising electrolyte additives or additive combinations on the different types of pouch cells. It is hoped that this report can be used as a guide or reference for the wise choice of electrolyte additives in $\text{Li}[\text{Ni}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}]\text{O}_2/\text{graphite}$, $\text{Li}[\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}]\text{O}_2/\text{graphite}$ and $\text{Li}[\text{Ni}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}]/\text{graphite}$ pouch cells and also to show the shortcomings of particular positive electrode compositions.

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1. Introduction

Li-ion cells are widely used in numerous applications, from portable electronics to electrified vehicles. In order to meet the increasing demands of these applications, suitable electrode materials and electrolyte systems, which can lead to higher energy density, higher power and longer cycle life, have been developed during the past two decades [1–3].

$\text{Li}[\text{Ni}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}]\text{O}_2$ (NMC111) is a popular positive electrode material because of its low cost, low toxicity and low reactivity with

electrolyte at elevated temperatures in the presence of suitable additives [4]. Higher nickel content in NMC can increase specific capacity to a particular cut-off potential, which improves energy density to that cut-off potential. $\text{Li}[\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}]\text{O}_2$ (NMC532) [5] is a widely used alternative to NMC111 and $\text{Li}[\text{Ni}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}]\text{O}_2$ (NMC622) [6,7] is considered to be a promising higher energy density material.

In addition to the choice of electrode materials, electrolyte additives can extend the lifetime and also increase the energy density of cells by allowing high voltage operation. Some well-known electrolyte additives such as vinylene carbonate (VC) and prop-1-ene-1,3-sultone (PES), which can increase the lifetime of cells, have been studied by many researchers. Aurbach et al. [8] showed that VC can decrease the impedance of LiNiO_2 and LiMn_2O_4

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electrodes at room temperature. Xia et al. [9] showed that both PES and VC can improve the coulombic efficiency (CE) and charge end-point capacity slippage for NMC111/graphite cells operated up to 4.2 V, which suggested a longer lifetime. PES can also dramatically suppress gas production during formation and cycling [10]. Furthermore, new electrolyte additives (e.g. pyridine-boron trifluoride (PBF) and its derivatives [11,12]) and new additive combinations (e.g. 2% PES + 1% methylene methanedisulfonate (MMDS) + 1% tris(trimethylsilyl) phosphite (TTSPi), called “PES211” [13]) have been developed to improve NMC111/graphite and $\text{LiNi}_{0.42}\text{Mn}_{0.42}\text{Co}_{0.16}\text{O}_2$ (NMC442)/graphite cell performance up to 4.4 V and even to 4.5 V by controlling impedance growth and improving capacity retention during long-term cycling.

There are no references about the comparison of the effects of useful electrolyte additives on different NMC compositions. In this work, the effects of several promising electrolyte additives and additive combinations on NMC111/graphite, NMC532/graphite and NMC622/graphite pouch cells were systematically investigated and compared. The ultra high precision charger (UHPC) at Dalhousie University [14] was used to characterize the various chemistries during charge–discharge cycling to 4.2 V or 4.4 V. Electrochemical impedance spectroscopy (EIS), gas evolution measurements and long-term “cycle-store” testing were also carried out. The data in this paper can be used to select additives that yield improvements for a particular NMC grade and also identify shortcomings of one NMC grade compared to another. Electrolyte additives like PES, PBF and “PES211”, that were initially developed in studies of NMC111/graphite cells cycled to 4.2 V [15] were found to work well in cells of NMC532 and NMC622 cycled to 4.2 V. This is re-assuring because “PES211” was found to be very poor in high “nickel content” NMC811/graphite cells [16] which suggests an important interplay between positive electrode surface chemistry and the electrolyte additives. Another interesting finding is that NMC622 cells always generate more gas during formation to 4.4 V or continuous cycling to 4.4 V than cells of the other grades, regardless of the electrolyte additives selected.

1.1. Experimental

1.0 M LiPF_6 in ethylene carbonate (EC):ethyl methyl carbonate (EMC) (3:7 by weight, from BASF, water content was 12.1 ppm) was used as the control electrolyte. Electrolytes with additives were formulated by dissolving 1 wt% PBF [12] or 2 wt% VC (from BASF, 99.97%), or PES (from Lianchuang Pharmaceutical, 98.2%) into the control electrolyte. Electrolytes with 2% PES + 1% MMDS (from Tinci Materials Technology, 98.7%) + 1% TTSPi (Sigma–Aldrich, >95%) (“PES-211”) or 0.5% pyrazine di-boron trifluoride (PRZ) + 1% MMDS [11] were also studied in this work. The chemical structures of the electrolyte additives used in this work are shown in Refs. [12,13].

1.1.1. Pouch cells

Dry (no electrolyte) NMC111/graphite, NMC532/graphite and NMC622/graphite pouch cells (210 mAh) balanced for 4.6 V operation were obtained from Amperex Technology Limited (No.1, Xingang Road, Zhangwan Town, Jiaocheng District, Ningde City, Fujian Province, PRC, 352100). Table 1 shows the metal element analysis results of these cells using the Inductively Coupled Plasma (ICP) technique. The negative electrode to positive electrode capacity ratios (N/P ratios) of these cells operated up to 4.2 V and 4.4 V are displayed in Table 2.

All pouch cells were vacuum sealed without electrolyte in a dry room in China and then shipped to our laboratory in Canada. Before electrolyte filling, the cells were cut just below the heat seal and dried at 80 °C under vacuum for 12 h to remove any residual water.

Table 1

Summary of metal content analysis results for the positive electrodes of NMC111/graphite, NMC532/graphite and NMC622/graphite pouch cells measured using ICP-OES. The sum of the metal contents has been normalized to 2.0. The error in each value is estimated to be ± 0.02 .

Cell type	Element			
	Li	Ni	Mn	Co
	Content			
NMC111	1.01	0.33	0.33	0.33
NMC532	0.98	0.47	0.31	0.25
NMC622	0.98	0.60	0.21	0.21

Table 2

Calculated N/P ratios of NMC111/graphite, NMC532/graphite and NMC622/graphite pouch cells operated up to 4.2 V and 4.4 V. (The reversible capacity of the fully lithiated negative electrode is about 265 mAh).

	NMC111/G	NMC532/G	NMC622/G
4.2 V capacity (mAh)	195	200	195
4.2 V N/P ratio	1.36	1.32	1.36
4.4 V capacity (mAh)	204	210	215
4.4 V N/P ratio	1.3	1.26	1.23

Then the cells were transferred immediately to an argon-filled glove box for filling and vacuum sealing. All the pouch cells were filled with 0.9 g of electrolyte. After filling, cells were vacuum-sealed with a compact vacuum sealer (MSK-115A, MTI Corp.). First, cells were placed in a temperature box at $40. \pm 0.1$ °C where they were held at 1.5 V for 24 h, to allow for the completion of wetting. Then, the pouch cells (called type-A cells) for operation up to 4.2 V were charged at 10 mA (C/20) to 3.8 V while the pouch cells (called type-B cells) for operation up to 4.4 V were charged at 10 mA (C/20) to 3.5 V. After this step, all the type-A cells and type-B cells filled with control electrolyte were transferred and moved into the glove box, cut open to release any gas generated and then vacuum sealed. Then all the type-A cells were charged at 10 mA (C/20) to 4.2 V while all the type-B cells were charged at 10 mA (C/20) to 4.4 V. After this step, all the type-B cells were transferred and moved into the glove box, cut open to release any gas generated and then vacuum sealed again.

1.1.2. Ultra high precision cycling experiments

Selected cells were cycled using the ultra high precision charger (UHPC) at Dalhousie University [14] between 2.8 and either 4.2 V or 4.4 V at $40. \pm 0.1$ °C using currents corresponding to C/20 for 16 cycles.

1.1.3. Cycle-store experiments

Selected cells were first charged to 4.4 V and discharged to 2.8 V twice with a C/20 current. Then the cells were cycled between 2.8 and 4.4 V using a current corresponding to C/5. At the top of every charge (4.4 V), the cells were left open circuit for 24 h while their potentials were monitored. An E-One Moli Energy cycler was used for this experiment with the cells located in a temperature controlled box ($40. \pm 0.1$ °C).

1.1.4. Electrochemical impedance spectroscopy (EIS)

EIS measurements were conducted on all the pouch cells before and after cycling. Cells were charged or discharged to 3.8 V before they were moved to a $10. \pm 0.1$ °C temperature box. Alternating current (AC) impedance spectra were collected with ten points per decade from 100 kHz to 10 mHz with a signal amplitude of 10 mV. A Biologic VMP-3 was used to collect this data.

1.1.5. Determination of gas evolution in pouch cells

In-situ (dynamic) and *ex-situ* (static) gas measurements were used to measure gas evolution during formation and cycling. According to Archimedes' principle, the changes in the weight of a cell suspended in fluid, before, during and after testing are directly related to the volume changes by the change in the buoyant force. The change in mass of a cell, Δm , suspended in a fluid of density, ρ , is related to the change in cell volume, Δv , by

$$\Delta v = -\Delta m / \rho \quad \text{eq. 1}$$

Ex-situ measurements were made by suspending pouch cells from a fine wire “hook” attached under a Shimadzu balance (AUW200D). The pouch cells were immersed in a beaker of de-ionized “nanopure” water (18 M Ω) that was at 20 ± 1 °C for measurement. Before weighing, all cells were charged or discharged to 3.80 V.

In-situ measurements were made using the apparatus and procedure described in Ref. [17].

2. Results and discussion

Self et al. [10] showed that during the first charge of NMC/graphite Li-ion cells (formation cycles) there are two gas production steps, the first step around 3.7 V is attributed to reactions at the negative electrode surface while the second gas step around 4.3 V is caused by reactions happening at the positive electrode. Fig. 1 shows a summary of the volume of gas produced in these two distinctive gas production steps of the different NMC/graphite pouch cells. The cells were studied with the in-situ gas generation apparatus [17] and contained control electrolyte and electrolyte with 2% VC or 2% PES. Two cells were measured for each data point and the error bar represents the standard deviation between the data. The amplitude of each step is strongly dependent on

temperature and electrolyte composition. In all cases, higher temperatures caused larger volume changes at both steps. 2% VC and 2% PES, especially 2% PES, suppress gas production at the negative electrode (the first gas step) at all temperatures. At the positive electrode (the second gas step) 2% PES decreased the amount of gas compared to control electrolyte while 2% VC increased gas production, especially at 70 °C where the amount of gas could not be measured accurately (indicated by the question mark shown in Fig. 1f) because cells with over ~ 3.5 mL of gas floated. The volume of gas produced during the second step increased with the increase of nickel content in NMC, which suggests the central role of nickel in gas production at the positive electrode of NMC/graphite cells. Pouch cell manufacturers contemplating NMC622 should be aware of this.

Fig. 2 shows the results of UHPC testing on the various NMC/graphite cell types containing different electrolyte additives tested to either 4.2 V (left panels) or 4.4 V (right panels). Fig. 2a and b show the fractional capacity fade per cycle, 2c and 2d show the fractional charge endpoint capacity slippage per cycle, while 2e and 2f show the coulombic efficiency (CE). The UHPC data were measured using a constant current corresponding to C/20 between 2.8 V and 4.2 V or 4.4 V at 40 °C. Each result in Fig. 2 is the average for two nominally identical cells except for data points where only one cell was available.

Based on the lithium inventory model reported by Smith et al. [18], the fractional fade per cycle (Fig. 2a and b), fade/Q, represents lithium loss due to SEI growth at the negative electrode and the fractional charge endpoint capacity slippage per cycle, slippage/Q, indicates electrolyte oxidation at the positive electrode. The relationship between fade/Q, slippage/Q and CE can be described in the following equation:

$$1 - \text{CE} = \text{coulombic inefficiency (CIE)} = \text{Fade/Q} + \text{Slippage/Q} \quad \text{eq. 2}$$

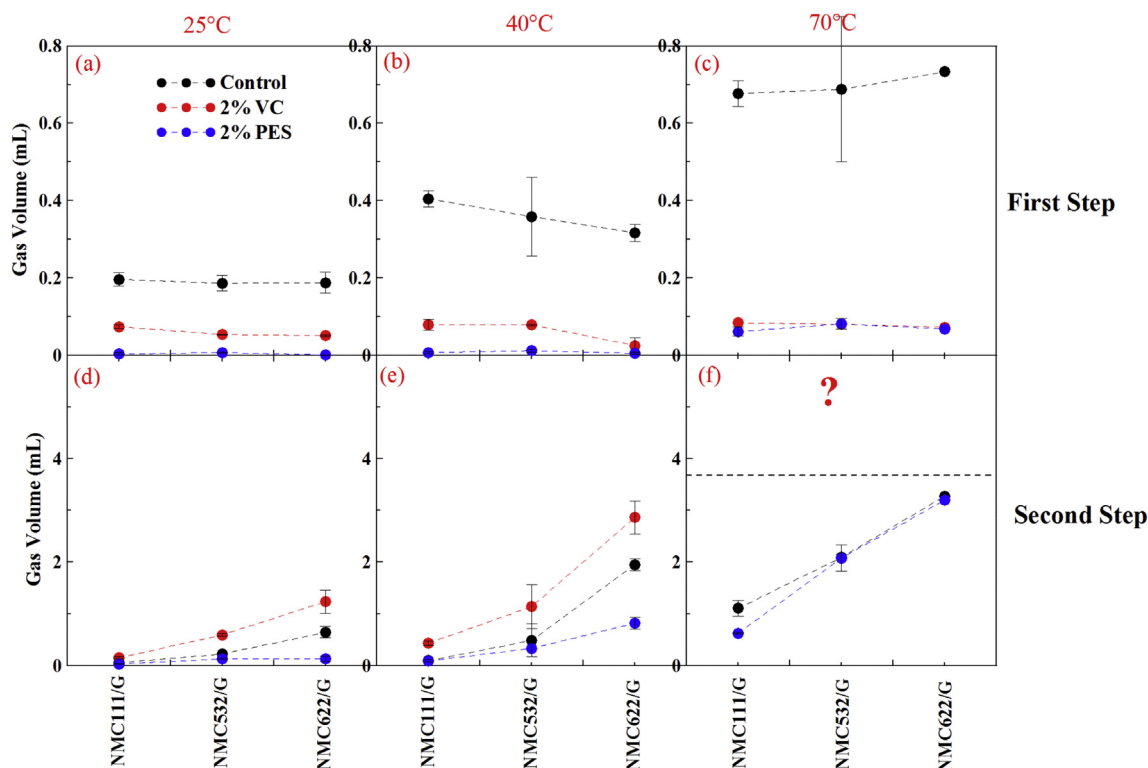


Fig. 1. Volume change during the negative electrode gas step (mostly ethylene – top panel) and during the positive electrode gas step (mostly CO₂ – bottom panels) for the three types of NMC/graphite pouch cells with control electrolyte, 2% VC and 2% PES at different temperatures as indicated. The top panel shows the volume change for cells at ~3.7 V, i.e. the first gas step [10]. The bottom panel shows the difference between the maximum gas produced at 4.6 V and the volume preceding this second step [10].

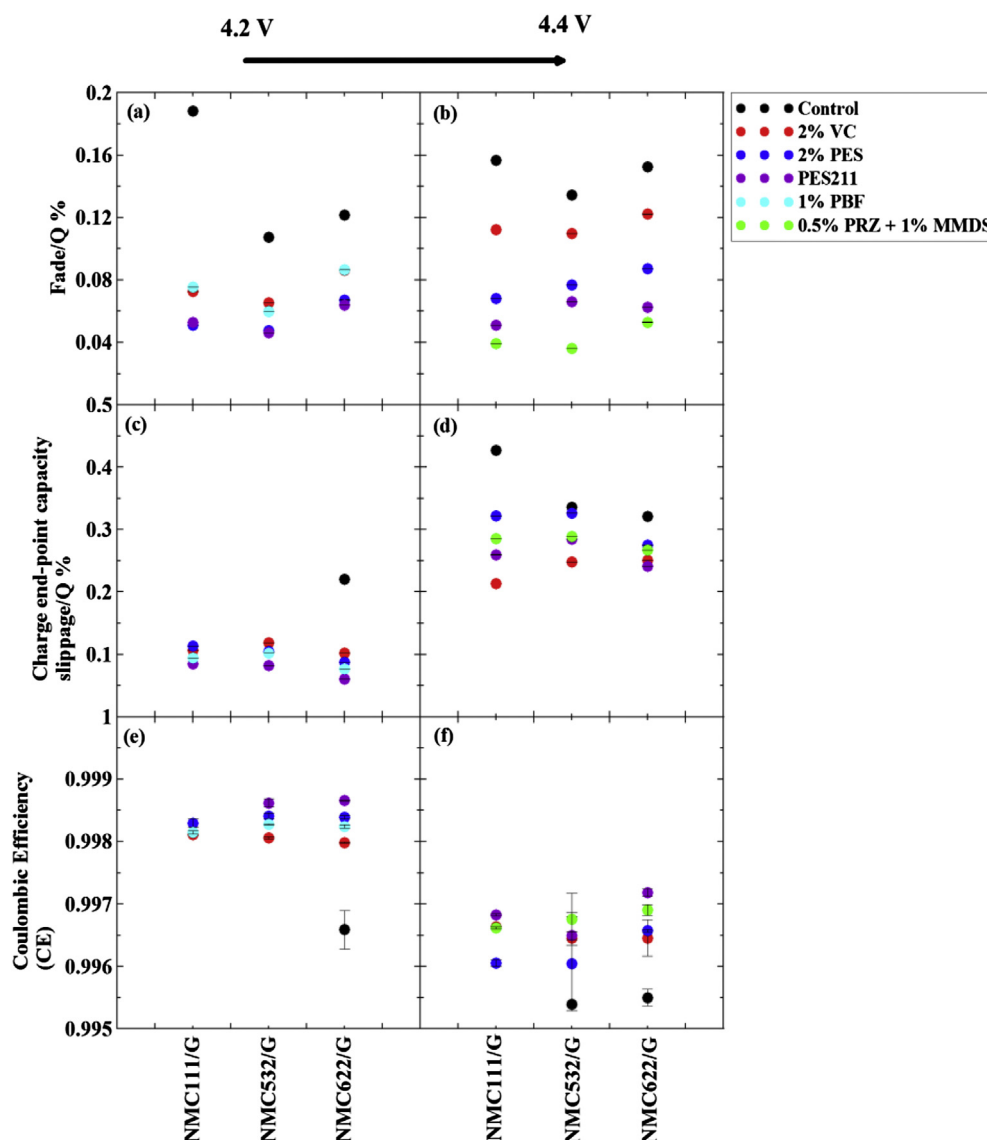


Fig. 2. Summary of important parameters for the three types of NMC/graphite pouch cells with various electrolyte additives during UHPC testing with a current of C/20 between 2.8 and 4.2 V (a, c and e) or 4.4 V (b, d and f) at 40 °C. Fig. 2a and b show the fractional fade per cycle, Fig. 2c and d show the fractional charge endpoint capacity slippage per cycle, while Fig. 2e and f show the coulombic efficiency. All the results were calculated using data from cycles 11 to 15.

Lower slippage/Q, lower fade/Q and higher CE (or lower CIE) generally result in longer lived cells [19].

The calculated CE from Eq. (2) and the measured CE are shown in Table 3. The calculated CE matches the measured CE which indicates the UHPC measurements are accurate.

Fig. 2a, c and e show fade/Q, slippage/Q and CE for the different NMC cells with different electrolytes, respectively, when the upper cut-off voltage is 4.2 V. As indicated in Table 3, some of the data for the control cells is off scale in Fig. 2 due to very poor performance. Wang et al. [15] showed that “PES211” yielded excellent performance in NMC111 cells charged to 4.2 V and “PES211” also yields the highest CE, the lowest slippage and the lowest fade for NMC532 and NMC622 cells. Nie et al. [12] showed that PBF was an effective additive for NMC442 and NMC111 cells which performed better than VC. Fig. 2a, c and e show that PBF outperforms VC for NMC532 and NMC622 demonstrating the wide applicability of PBF.

Fig. 2b, d and f show the UHPC results when the upper cut-off voltage was increased to 4.4 V. When the cells are cycled to 4.4 V instead of 4.2 V, the charge endpoint capacity slippage increased

dramatically (compare Fig. 2d to c) and the coulombic efficiency decreased dramatically (compare Fig. 2f to e). This indicates that a dramatic increase in the rate of electrolyte oxidation occurs between 4.2 V and 4.4 V for all of NMC111, NMC532 and NMC622 cells. The capacity loss per cycle for control and 2% VC increases significantly from 4.2 V (Fig. 2a) to 4.4 V (Fig. 2b) while the increase is minimal for “PES211”, suggesting this additive blend is quite effective for all the NMC grades tested here. Cells with “PES211” or 0.5% PRZ + 1% MMDS have the highest CE for all NMC grades when cycling is to 4.4 V. The CE is highest for NMC622 cells with “PES211” and it will be shown later that such cells have the best long term capacity retention.

Fig. 3 summarizes the amount of gas evolved in NMC/graphite cells tested to 4.2 V after formation (Fig. 3a) and after UHPC cycling (Fig. 3b) as well as for cells tested to 4.4 V during formation (Fig. 3d) and cycling (Fig. 3e). Fig. 3a shows that all cells with control electrolyte and with 1% PBF generate about 0.2 mL of gas (the original cell volume is 2.2 ml) during formation to 4.2 V while cells with 2% VC, 2% PES or “PES211” generate virtually no gas. Fig. 3b shows that

Table 3

Summary of measured CE during UHPC test between 2.8 and 4.2 V or 4.4 V at 40 °C with a current of C/20 and calculated CE based on Eq. (2) for three types of NMC/graphite cells containing different electrolyte additives or additive combinations.

		Control		2%VC		2% PES	
		CE _{Mea}	CE _{Cal}	CE _{Mea}	CE _{Cal}	CE _{Mea}	CE _{Cal}
4.2 V	NMC111/G	0.9874	0.9877	0.9981	0.9982	0.9983	0.9984
	NMC532/G	0.9928	0.9931	0.9981	0.9982	0.9984	0.9985
	NMC622/G	0.9964	0.9966	0.9980	0.9981	0.9984	0.9984
4.4 V	NMC111/G	0.9940	0.9942	0.9966	0.9967	0.9960	0.9961
	NMC532/G	0.9954	0.9953	0.9964	0.9964	0.9959	0.9960
	NMC622/G	0.9954	0.9953	0.9965	0.9963	0.9966	0.9964
4.2 V		PES211		1% PBF		0.5% PRZ + 1% MMDS	
	NMC111/G	0.9986	0.9986	0.9982	0.9983	N/A	N/A
	NMC532/G	0.9987	0.9987	0.9983	0.9984	N/A	N/A
4.4 V	NMC111/G	0.9968	0.9969	N/A	N/A	0.9966	0.9968
	NMC532/G	0.9965	0.9965	N/A	N/A	0.9969	0.9967
	NMC622/G	0.9971	0.9970	N/A	N/A	0.9970	0.9968

all cells generate virtually no gas during the 600 h of UHPC cycling between 2.8 and 4.2 V at 40 °C, regardless of the NMC grade or the additives selected. Fig. 3d shows that when cells were formed to 4.4 V, cells with control electrolyte and cells with 2% VC generated more than 0.1 mL of gas and of these cells, cells with NMC622 generated the most gas. Fig. 3e shows that after 600 h of UHPC cycling to 4.4 V at 40 °C, cells with 2% VC produced a substantial amount of gas which increased with the Ni content in NMC. In fact, even for the cells containing PES or PRZ, which generated very little gas, the amount of gas was largest for NMC622.

Fig. 3c and f show the impedance of all the NMC/graphite pouch cells measured after formation and then after the 600 h UHPC

cycling test to either 4.2 V and 4.4 V, respectively. The impedance reported is the diameter of the semi-circle of the Nyquist plot which predominantly represents the sum of the charge-transfer resistances, R_{ct} , at both the positive and negative electrodes. The impedance values are very similar for cells of different NMC grades with the same electrolyte additives except for the impedance before cycling of cells with 2% PES. Fig. 3c and f show that cells with control electrolyte or electrolyte with 2% VC increase their impedance after cycling while the other electrolyte additives or additive combinations decrease their impedance after cycling for both 4.2 V and 4.4 V testing.

In order to better distinguish the effects of the selected electrolyte additives on the NMC111, NMC532 and NMC622/graphite pouch cells at 4.4 V, an aggressive “cycle-store” protocol was used to expose cells to high potential for significant fractions of their testing time (24 h each cycle). Fig. 4 shows the discharge capacity versus cycle number for all cells tested using the “cycle-store” protocol at 40 °C. Fig. 4a shows that the nickel content does not change the cycling performance with control electrolyte as all cells decrease to ~180 mAh capacity (~85% of the cell capacity) after ~25 cycles. Fig. 4b shows that 2% VC does not improve matters significantly for NMC111/graphite cells but cells with NMC532 or NMC622 last 40 ± 7 or 55 ± 5 cycles to 180 mAh, respectively. Fig. 4c shows that the cycle lives to 180 mAh for NMC111, NMC532 and NMC622 cells with 2% PES are 40 ± 2 , 40 ± 2 and 45 ± 6 cycles. Fig. 4d shows that the cycle lives to 180 mAh for NMC111, NMC532 and NMC622 cells with “PES211” are 53 ± 2 , 53 ± 2 and 85 ± 5 (extrapolated) cycles, respectively. Fig. 4e shows that the cycle lives to 180 mAh for NMC111, NMC532 and NMC622 cells with 0.5% PRZ + 1% MMDS are 65 ± 3 , 65 ± 3 and 57 ± 8 cycles, respectively. NMC622 cells with “PES211” perform best in this aggressive “cycle-store” test at 40 °C up to 4.4 V. This is as expected based on the CE results in Fig. 2f.

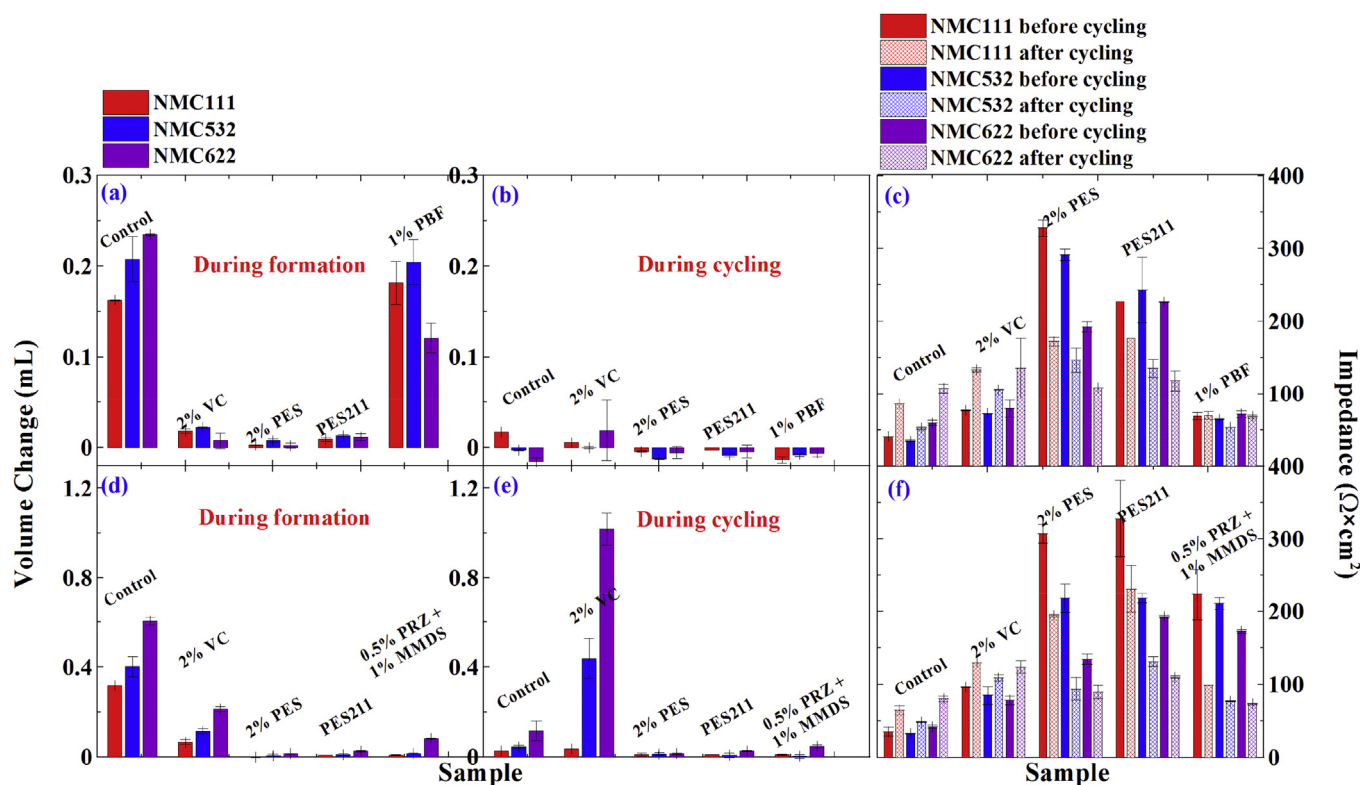


Fig. 3. Summary of gas evolution for the three types of NMC/graphite pouch cells with various selected electrolyte additives or additive combinations tested using the UHPC between 2.8 and 4.2 V during formation (a) and cycling (b) and for cells tested between 2.8 and 4.4 V during formation (d) and cycling (e). Summary of impedance (R_{ct}) after formation and after UHPC cycling between 2.8 and 4.2 V (c) or 4.4 V (f).

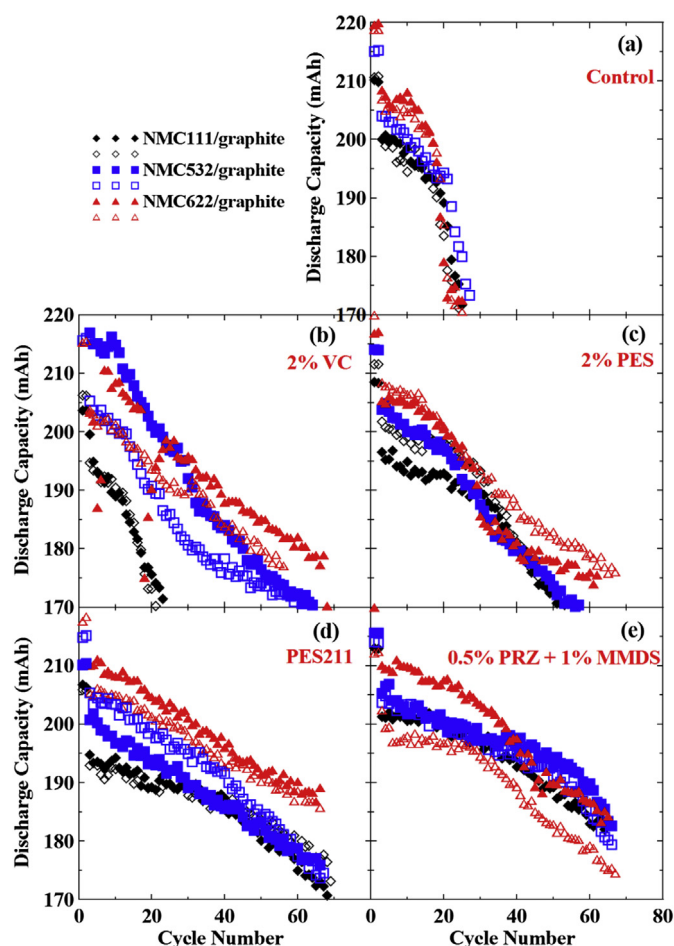


Fig. 4. Discharge capacity versus cycle number for the three types of NMC/graphite pouch cells during “cycle-store” testing between 2.8 and 4.4 V at 40 °C using different electrolytes: (a) control (b) 2% VC (c) 2% PES (d) “PES211” and (e) 0.5% PRZ + 1% MMDS.

Fig. 5 shows a summary of the voltage decay during the storage period of the “cycle-store” experiments detailed in Fig. 4. The differences in voltage decay from cell to cell result from the parasitic reactions at the positive electrode (e.g. electrolyte oxidation) and also from differences in direct current (DC) cell resistance. The latter effect is observed by a rapid potential drop when the cells switch from charge to open circuit. Fig. 5a, b and c show the potential versus time during the 35th storage period for NMC111, NMC532 and NMC622/graphite cells, respectively. Fig. 5a, b and c show that cells with control electrolyte develop large DC resistance. NMC111 and NMC532 cells with 2% VC develop a serious self-discharge while cells with NMC622 and 2% VC do not. By contrast, all cells with “PES211” and with 0.5% PRZ + 1% MMDS, show small voltage variation during the storage period and small values of the voltage change between the beginning and end of the storage period, which we call “ V_{drop} ” here. Fig. 5d, e and f show a summary of V_{drop} for all cells at cycles 1, 15 and 35, respectively. Fig. 5d and e show that, apart from NMC111 cells with 2% VC, V_{drop} is relatively stable over the first 15 cycles. Fig. 5f shows that V_{drop} increases strongly for all cells with control electrolyte and for NMC111 and NMC532 cells with 2% VC between cycle 15 and cycle 35. All other cells show relatively stable values of V_{drop} over the first 35 cycles, especially those with “PES211” and 0.5% PRZ + 1% MMDS.

Fig. 6a shows the volume of gas evolved in the NMC/graphite pouch cells with the various electrolyte additives after the “cycle-store” test described by the results in Figs. 4 and 5. Fig. 6a is very

interesting because the trends in Fig. 6a do not match those in Fig. 3. In particular, cells with “PES211” and cells with 0.5% PRZ + 1% MMDS show significant gas production compared to other cells in Fig. 6, but the same cells show much less gas production compared to other cells in Fig. 3. This can be understood by considering Fig. 5a, b and c, where all cells with “PES211”, 0.5% PRZ + 1% MMDS and NMC622 cells with 2% VC remain at high potential during the repeated 24 h storage periods, hence are exposed to more oxidizing conditions than cells with control electrolyte, for example. NMC622 cells containing “PES211” or 0.5% PRZ + 1% MMDS generate more gas than NMC532 and NMC111 cells with the same additives. Cells with only 2% PES generate very little gas in this test even though their potential remains high during the storage periods. This is because of the excellent ability of PES itself to suppress gas evolution.

Fig. 6b shows the impedance (R_{ct}) for the pouch cells described by Figs. 4 and 5 measured after formation and after the “cycle-store” testing to 4.4 V at 40 °C. After the cycle testing, the cells were degassed prior to measuring the impedance at 3.8 V and 10 °C. Due to the long exposure time to high potential, the impedance of most of the cells increased after cycling except for NMC111/graphite and NMC532/graphite cells filled with 0.5% PRZ + 1% MMDS, which suggests that this additive combination is most suitable, of the ones studied, for controlling impedance growth.

One of the reasons for undertaking the work in this paper was to investigate whether the “PES211”, PBF and PRZ + MMDS additives that were developed using NMC111/graphite and NMC442/graphite pouch cells were also effective in NMC532 and NMC622 cells. The results in this paper show that these additives work well for all these NMC grades although gas generation during formation and testing to 4.4 V for NMC622 is more problematic. One question that remains is: Is there a combination of NMC grade and electrolyte additive that is “best” for cells operated to 4.2 V or to 4.4 V?

Fig. 7 shows “radar” plots that compare the effects of selected electrolyte additives or additive combinations on NMC111/graphite (Fig. 7a), NMC532/graphite (Fig. 7b) and NMC622/graphite (Fig. 7c) pouch cells during the UHPC cycling to 4.2 V. The three axes in the radar plots represent the average coulombic inefficiency (CIE) (from 11 to 15 cycles), the average charge end-point capacity slippage (from 11 to 15 cycles) and R_{ct} after UHPC cycling. The value in brackets at the end of each axis is the maximum value the axis. The axes have been scaled so that 100% is the value of the additive that has the largest (the worst) value of each parameter. [Control electrolyte is so poor in NMC111 and NMC532 cells that it has been omitted from Fig. 7a and b.] Therefore the best additive would have values closest to the center of the plot. Fig. 7 shows that there are trade-offs that can be made when selecting additives, but that 1% PBF and “PES211” are most interesting. For example, 1% PBF, shows the lowest impedance after cycling and a comparable CIE and slippage to 2% VC. “PES211” shows the best CIE and the lowest slippage, but has higher impedance than 1% PBF.

Fig. 7 also shows radar plots that compare the three NMC grades when used with the 1% PBF additive (Fig. 7d) and when used with “PES211” additive blend (Fig. 7e). The results in Fig. 7d and e are for the UHPC cycling to 4.2 V described by Fig. 2. The results here suggest that the longest lifetime cells using these additive blends will be NMC622/graphite cells, however these additives are also very effective for NMC111 and NMC532. Fig. 3 shows that gas generation with these additives is not problematic for cycling to 4.2 V at 40 °C. The radar plots in Fig. 7d and e run from 60% to 100% in order to make the comparisons more facile.

Fig. 8 shows a “radar” plot which compares the effects of selected electrolyte additives or additive combinations on NMC111/graphite (Fig. 8a), NMC532/graphite (Fig. 8b) and NMC622/graphite (Fig. 8c) pouch cells studied using both UHPC results (4.4 V –

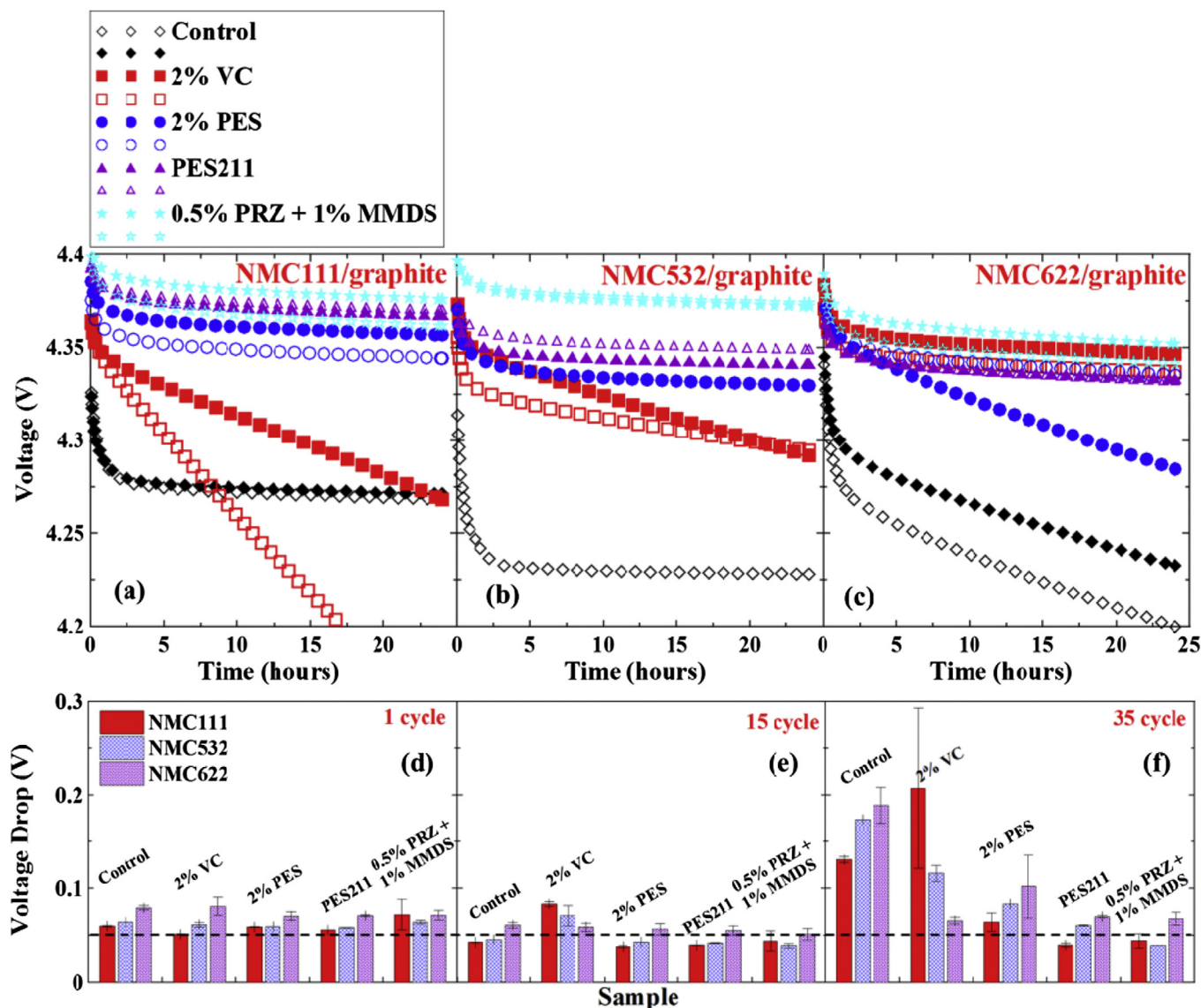


Fig. 5. Summary of the voltage drop (V_{drop}) for three types of NMC/graphite pouch cells with various electrolyte additives during "cycle-store" testing between 2.8 and 4.4 V at 40 °C after a certain number of cycles: 1 cycle (d), 15 cycles (e) and 35 cycles (f). Voltage versus time for NMC111/graphite (a), NMC532/graphite (b) and NMC622/graphite (c) cells with various electrolyte additives or additive combinations at cycle 35 during the storage period of "cycle-store" testing.

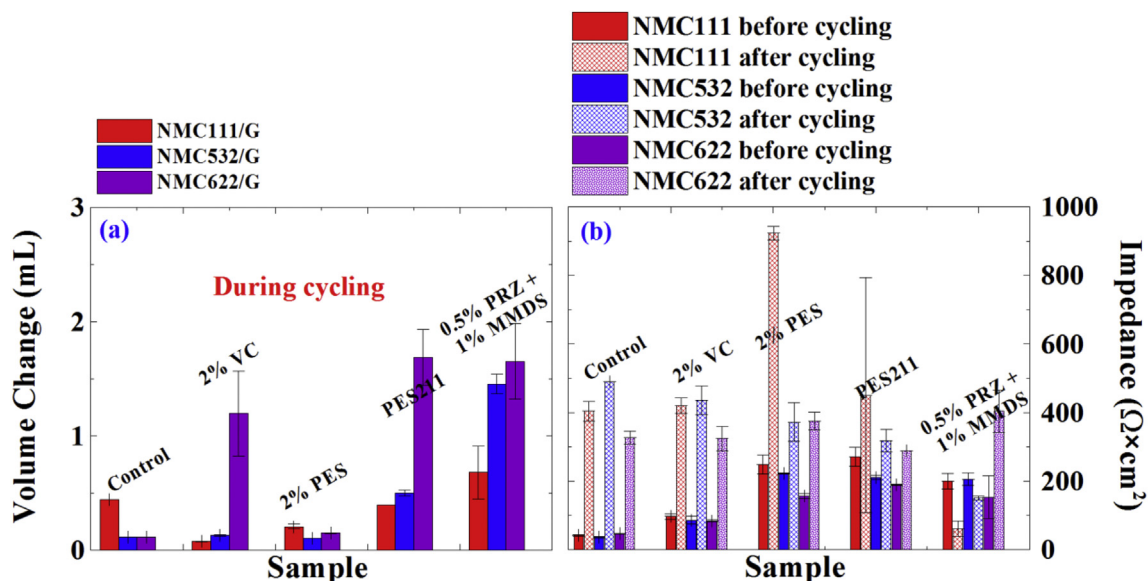


Fig. 6. Summary of gas evolution (a) and impedance before and after "cycle-store" testing (b) between 2.8 and 4.4 V at 40 °C for the three types of NMC/graphite pouch cells with various electrolyte additives.

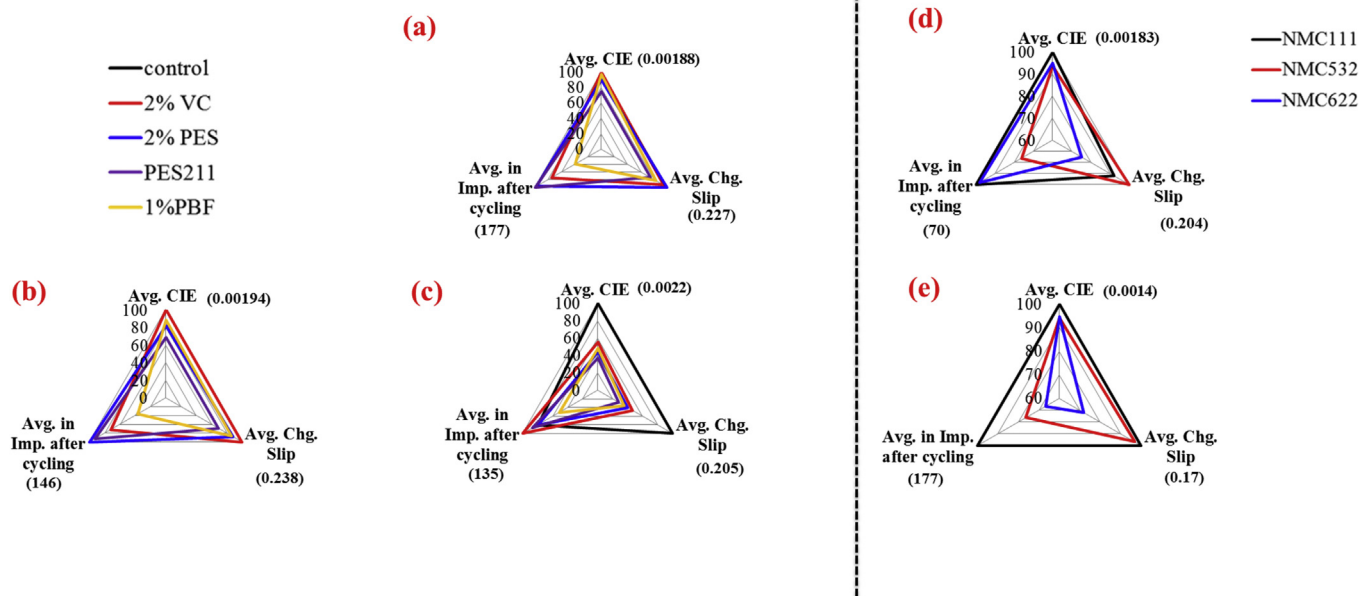


Fig. 7. Radar plots summarizing the effects of selected electrolyte additives or additive combinations on NMC111/graphite (a), NMC532/graphite (b) and NMC622/graphite (c) pouch cells studied using UHPC up to 4.2 V. The axes are normalized to the worst value being equal to 100% and the axes are: coulombic inefficiency ($CIE = 1 - CE$), charge end point capacity slippage and the value of R_{ct} after cycling. Radar plots summarizing the effects of 1% PBF (d) and "PES211" (e) on NMC111/graphite, NMC532/graphite and NMC622/graphite pouch cells studied using the UHPC up to 4.2 V. These radar plots run from 60% to 100% of the maximum value. Each data point in Fig. 7 represents the average of two cells. Values closest to the center of the radar plot are best.

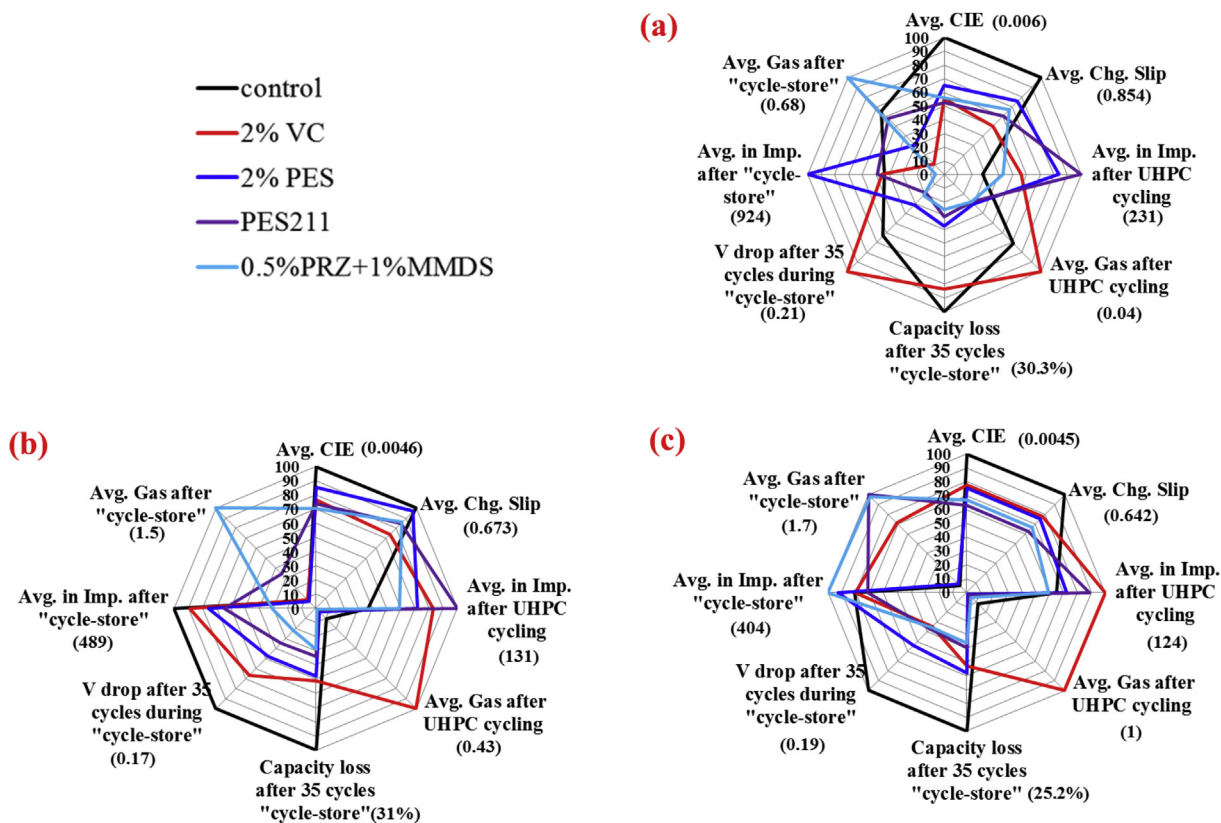


Fig. 8. Radar plots summarizing the effects of selected electrolyte additives (combinations) on NMC111/graphite (a), NMC532/graphite (b) and NMC622/graphite (c) pouch cells studied using UHPC and the "cycle-store" procedure to 4.4 V. Each data point in Fig. 8 represents the average of two cells. The axes are normalized to the worst value being equal to 100% and they consist of the average coulombic inefficiency (CIE) (from 11 to 15 cycles), the average charge end-point capacity slippage (from 11 to 15 cycles), the impedance (R_{ct}) after UHPC cycling, the gas evolution during UHPC cycling, the capacity loss after 35 "cycle-store" cycles, the impedance after the whole "cycle-store" process, the voltage drop at 35 "cycle-store" cycle and the gas evolution during the whole "cycle-store" process. Values closest to the center of the radar plot are best.

Fig. 2b, d, f and Fig. 3d, e and f) and the “cycle-store” procedure to 4.4 V (Figs. 4–6). The eight axes in the radar plots represent the average coulombic inefficiency (CIE) (from 11 to 15 cycles), the average charge end-point capacity slippage (from 11 to 15 cycles), the impedance after UHPC cycling, the gas evolution during UHPC cycling, the capacity loss after 35 “cycle-store” cycles, the impedance after the whole “cycle-store” process, the voltage drop after 35 “cycle-store” cycles and the gas evolution after the whole “cycle-store” process. The maximum value of each parameter is shown in brackets under each axis label of the radar plot. Although there is no absolute “winner” among the selected electrolyte additives or combinations it is clear that “PES211” and 0.5% PRZ + 1% MMDS yield significant advantages in most aspects of cell performance (e.g. CE, impedance control, etc.) compared to the other electrolyte additives.

3. Conclusions

This report has shown high precision cycling, EIS data, gas evolution measurements and “cycle-store” data for NMC111/graphite, NMC532/graphite and NMC622/graphite pouch cells containing selected electrolyte additives or additive combinations. The UHPC data showed a small difference in lifetime expectations between different NMC/graphite cells when the same additives were used. “PES211”, PBF and PRZ + MMDS additives are very effective in cells of all NMC grades. Although NMC622 can increase the energy density of cells charged to 4.4 V, it results in more gas production during formation and cycling, even with the best additives explored here. The reactivity of the various charged NMC electrode materials with electrolytes at elevated temperature from a safety perspective has been considered in another publication [20].

It will take many man-years of effort to elucidate the mechanisms of action of the various electrolyte additives and blended additives used here. Some details of the way that the additive PES functions can be found in the recent paper by Self et al. [21]. Further work on the “PES211” blended additive is in progress.

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